

KeyJoint-STGCN: Keypoint-Based Spatio-Temporal Graph Learning for Monocular Trajectory Prediction

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Abstract

Autonomous driving systems require robust motion understanding capabilities to ensure safe navigation in highly dynamic traffic environments. Accurate trajectory prediction of pedestrians and vehicles from monocular vision remains challenging due to viewpoint variations, occlusions, heterogeneous interactions, and complex scene dynamics. This paper presents KeyJoint-STGCN, a keypoint-driven spatio-temporal graph learning framework for monocular trajectory prediction in traffic scenes. The proposed approach first employs OpenPifPaf to estimate and track semantic keypoints of pedestrians and vehicles from monocular video sequences. These keypoints are then represented as graph nodes enriched with spatial position and temporal velocity features. A Spatio-Temporal Graph Convolutional Network (STGCN) is subsequently used to model motion dynamics and forecast future trajectories. To capture heterogeneous interactions among multiple road entities within the same frame, the framework constructs a multigraph representation with multiple edge connections encoding spatial and temporal dependencies at the keypoint level. The proposed method is evaluated on publicly available autonomous driving datasets, including JAAD, nuScenes, and BDD100K. Experimental results demonstrate that the proposed keypoint-centric graph representation improves trajectory forecasting performance and generalization capability compared with existing state-of-the-art trajectory prediction methods. The proposed framework highlights the effectiveness of combining semantic keypoint reasoning with spatio-temporal graph learning for intelligent transportation and computer vision applications.

Keywords: Keypoint-based representation, Monocular vision, Multi-agent interaction, Spatio-temporal graph convolution, Trajectory prediction.

1 Introduction

Trajectory prediction is an important research problem in computer vision and intelligent transportation systems due to its applications in autonomous driving, Advanced Driver Assistance Systems (ADAS), and traffic scene understanding [Ratre and Joshi, 2025]. Predicting the future motion of pedestrians and vehicles enables autonomous agents to better interpret dynamic environments and improve decision-making in complex urban traffic scenarios. However, accurate trajectory forecasting remains challenging because of behavioural uncertainty, occlusions, environmental variability, and heterogeneous interactions among multiple road users,

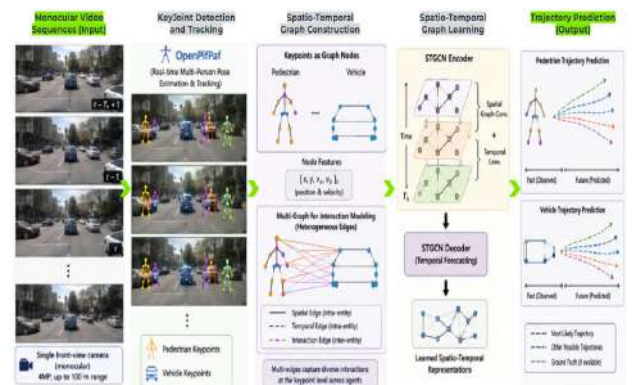


Figure 1: Overview of Proposed KeyJoint-STGCN

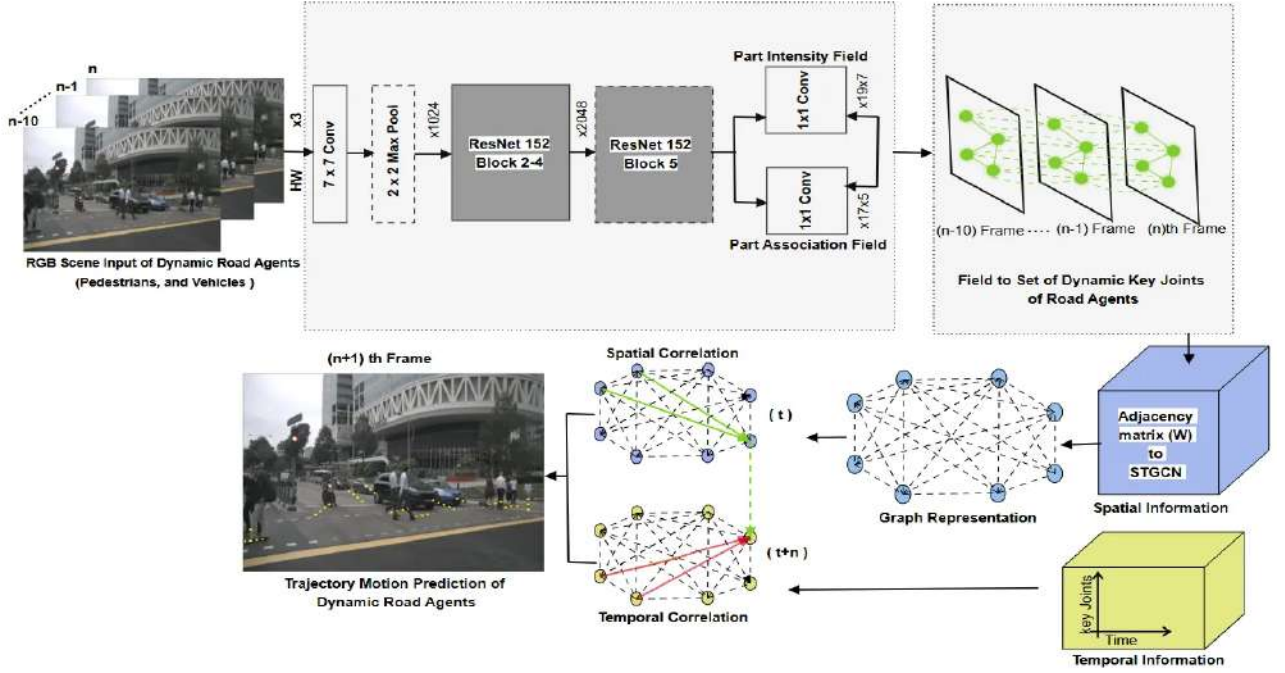


Figure 2: Proposed KeyJoint-STGCN Framework: The key joints detection process is done using the OpenPifPaf method, and the subsequent construction of a spatio-temporal graph for trajectory prediction. The graph is further enhanced using a multigraph approach. The STGCN is then employed to model temporal and spatial dependencies efficiently.

particularly in monocular vision settings briefed in Fig. 1 where depth perception is limited [Karle et al., 2022].

Recent deep learning approaches, including Social-GAN and Social-STGCNN [Xie et al., 2024], have demonstrated promising performance by modeling temporal dependencies and social interactions among agents. Similarly, graph-based methods such as GNNs and STGCNs have shown strong capability in learning spatial and temporal relationships in trajectory prediction tasks [Hsu and Chen, 2025]. Nevertheless, most existing approaches rely on coarse object-level representations, such as centroids or bounding boxes, and primarily focus on pedestrian prediction. As a result, they often fail to capture fine-grained articulated motion dynamics and semantic interaction cues that are critical in complex traffic environments involving heterogeneous road entities [Luo et al., 2024].

Pose and keypoint estimation methods, including OpenPose, and OpenPifPaf [Lambricht et al., 2025], provide expressive representations of articulated motion and have shown effectiveness in action recognition and pedestrian intention analysis. However, semantic keypoint information remains underexplored in trajectory prediction frameworks, particularly for multi-class road-user interaction modeling in monocular driving scenarios.

To address these limitations, this paper proposes *KeyJoint-STGCN*, a keypoint-based spatio-temporal graph learning framework for monocular trajectory prediction. The proposed method employs OpenPifPaf [Kreiss et al., 2021] to estimate and track 17 human body keypoints and 24 vehicle semantic keypoints (via the pre-trained Apollo-Car3D model checkpoint) from monocular video sequences. These keypoints are represented as graph nodes with position and velocity features, while a multigraph interaction strategy is introduced to model heterogeneous spatial and temporal dependencies among road entities at the keypoint level. Subsequently, a Spatio-Temporal Graph Convolutional Network (STGCN) is utilized to learn motion dynamics and forecast future trajectories [Kahl et al., 2026].

An overview of the proposed framework is illustrated in Fig. 2. Unlike existing approaches that depend on multi-sensor fusion or bird’s-eye-view representations, the proposed framework operates entirely on monocular

front-view camera input, reducing computational complexity while maintaining efficient motion representation capability. The proposed method is evaluated on publicly available autonomous driving datasets, namely JAAD, nuScenes, and BDD100K [Kotseruba et al., 2016, Caesar et al., 2020, Yu et al., 2020]. Experimental results demonstrate improved trajectory prediction accuracy and generalization capability compared with existing state-of-the-art methods.

The pivotal research contributions of this work are summarized as follows:

- A novel keypoint-centric STGCN framework for monocular pedestrian and vehicle trajectory prediction.
- A multigraph interaction modeling strategy for capturing heterogeneous spatial and temporal dependencies at the semantic keypoint level.
- An efficient monocular vision-based trajectory prediction pipeline without relying on multi-sensor fusion architectures.
- Extensive validation on benchmark autonomous driving datasets demonstrating improved forecasting performance over state-of-the-art approaches.

2 Proposed Methodology: KeyJoint-STGCN

The proposed KeyJoint-STGCN framework consists of three major stages: semantic keypoint detection, spatio-temporal multigraph construction, and trajectory prediction using a Spatio-Temporal Graph Convolutional Network (STGCN). An overview of the framework is illustrated in Fig. 2.

2.1 Problem Formulation

Given M dynamic road entities observed over previous time steps $\{T_{n-10}, \dots, T_{n-1}\}$, the objective is to predict their future trajectories at time $T_{n+\Delta n}$. Each entity is represented using semantic key joints instead of coarse object-level centroids. For each key joint, the feature vector is defined as:

$$q_i = \{x_i, y_i, v_{x,i}, v_{y,i}\} \quad (1)$$

where (x_i, y_i) denote spatial coordinates and $(v_{x,i}, v_{y,i})$ represent velocity components.

2.2 Semantic Keypoint Detection

The proposed framework employs OpenPifPaf [Kreiss et al., 2022] for semantic keypoint tracking. Standard 17 COCO joints represent pedestrians, while a sparsified 24-keypoint vehicle skeleton (from ApolloCar3D) represents vehicle geometry. OpenPifPaf jointly estimates articulated key joints and establishes temporal associations across consecutive frames using Composite Fields and Temporal Composite Association Fields (TCAF). This enables robust tracking of pedestrians and vehicles without explicit data association.

2.3 Spatio-Temporal Multigraph Construction

For each frame, detected semantic key joints form multigraph nodes $G = (V, E)$. Horizontal, vertical, and velocity interactions are defined as

$$a_{ij}^d = \exp\left(-\alpha_d \frac{\|\mathbf{f}_i^d - \mathbf{f}_j^d\|^2}{N_d^2}\right), \quad d \in \{x, y, v\}, \quad (2)$$

where $\mathbf{f}^x = x$, $\mathbf{f}^y = y$, $\mathbf{f}^v = (v_x, v_y)$, and $N_x = W$, $N_y = H$, $N_v = V_{\max}$ denote the corresponding normalization factors. The fused adjacency matrix is

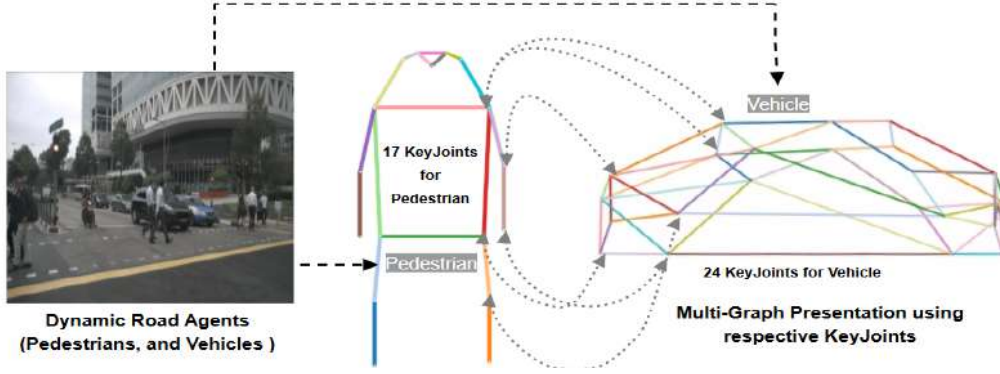


Figure 3: Graph construction process using KeyJoint-STGCN approach. Key joints are treated as individual nodes. Position and velocity data are incorporated into these nodes, forming the foundation of the graph. A multigraph is created to capture interactions between distinct entities, connecting one key joint (node) of one entity to another entity’s corresponding key joint (node) with multiple edges. As illustrated in Fig. 3, horizontal, vertical, and velocity edge relations are constructed via Eq. 2 and weighted by hyper-parameters $\lambda_x = 0.4, \lambda_y = 0.3, \lambda_v = 0.3$.

$$A = \lambda_x A_x + \lambda_y A_y + \lambda_v A_v, \quad \lambda_x + \lambda_y + \lambda_v = 1, \quad (3)$$

which is normalized before STGCN as

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}. \quad (4)$$

The resulting multigraph jointly models articulated keypoint structure and heterogeneous inter-agent interactions.

2.4 Trajectory Prediction using KeyJoint-STGCN

The constructed spatial graphs are fused into a unified interaction graph and processed using a Spatio-Temporal Graph Convolutional Network (STGCN) to jointly model spatial and temporal dependencies. The graph convolution operation is defined as:

$$H^{(l+1)} = \sigma(\bar{D}^{-1/2} \bar{A} \bar{D}^{-1/2} H^{(l)} W^{(l)}) \quad (5)$$

where $\bar{A}$ denotes the adjacency matrix with self-loops, $\bar{D}$ is the degree matrix, $H^{(l)}$ represents the node features, and $W^{(l)}$ denotes learnable weights.

Finally, a 3-layer Temporal Convolutional Network (TCN) with residual connections (kernel size $k = 3$, dilation factors $d \in \{1, 2, 4\}$, hidden channel size 64) forecasts future coordinates over the target horizon.

3 Experimental Results

3.1 Experimental Setup

The KeyJoint-STGCN framework is implemented in PyTorch and evaluated on a system with an Intel i7-13850HX CPU, 64 GB RAM, and a 4 GB GPU. Training uses the Adam optimizer with a learning rate of 1×10^{-4} and batch size of 128, converging in 50 epochs. Performance is assessed using ADE, FDE, and MSE for trajectory prediction accuracy.

Experiments are conducted on three benchmark datasets: JAAD (pedestrian behavior), nuScenes (multi-agent trajectory), and BDD100K (diverse urban driving) [Kotseruba et al., 2016, Caesar et al., 2020, Yu et al., 2020]. JAAD primarily focuses on pedestrian behaviour in urban driving scenarios, while nuScenes and BDD100K

provide large-scale multi-class traffic scenes with diverse environmental conditions and dynamic interactions. These datasets enable comprehensive evaluation of monocular trajectory prediction across pedestrians and vehicles.

3.2 Training and Validation Strategy

For all datasets, videos are divided into training, validation, and testing subsets. The proposed framework uses 10 observed frames to predict future trajectories over a 3-second horizon. OpenPifPaf is employed for semantic keypoint detection and tracking, and the extracted keypoints are converted into spatio-temporal multigraph representations before STGCN-based learning.

To validate model robustness, additional experiments are conducted using different pose trackers, including AlphaPose and PoseFlow [Wu et al., 2025]. Furthermore, ablation analysis is performed by varying the number of input frames to evaluate temporal dependency learning.



Figure 4: Examples of semantic keypoint detection for pedestrians and vehicles using OpenPifPaf on JAAD and BDD100K datasets.

3.3 Quantitative Performance Comparison

The proposed KeyJoint-STGCN is compared with state-of-the-art trajectory prediction methods, including DESIRE [Lee et al., 2017], Semantic-STGCNN [Rainbow et al., 2021], and GATraj [Cheng et al., 2023]. Tables 1 summarize the quantitative results on JAAD, nuScenes, and BDD100K datasets.

Table 1: Comprehensive performance comparison of trajectory prediction methods across JAAD, nuScenes, and BDD100K datasets.

Method	JAAD			nuScenes			BDD100K		
	ADE↓	FDE↓	MSE↓	ADE↓	FDE↓	MSE↓	ADE↓	FDE↓	MSE↓
DESIRE	0.8123	0.9347	703	2.4896	4.5923	1476	1.8901	2.9876	1075
Semantic-STGCNN	0.6975	0.7489	587	2.3128	4.2789	1201	1.5678	2.3789	823
GATraj	0.6450	0.7712	510	2.1050	4.2501	1035	1.4502	2.1405	755
KeyJoint-STGCN	0.6145	0.7214	462	2.1035	4.1107	932	1.2345	1.9456	741

Across all datasets, the proposed method consistently achieves lower ADE and FDE compared with existing methods. The improvements demonstrate the effectiveness of semantic keypoint-level motion representation and multigraph interaction modeling for monocular trajectory prediction.

3.4 Class-wise and Ablation Analysis

To further analyze performance across road-user categories, class-wise evaluation is conducted on the JAAD dataset. While pedestrian gains are substantial, vehicle trajectory forecasting is highly competitive with GATraj (1.9843 vs 1.9820 ADE on JAAD). This slight difference occurs because rigid vehicle dynamics are already well-approximated by bounding box centroids, unlike highly dynamic, articulated pedestrian body movements.

Table 2: Class-wise evaluation and pose tracker comparison on JAAD dataset.

Class-wise Evaluation				Pose Tracker Evaluation		
Class	Method	ADE↓	FDE↓	Pose Tracker	ADE↓	FDE↓
Vehicle	GATraj	1.9820	3.8862	PoseFlow	0.5446	0.6951
	KeyJoint-STGCN	1.9843	3.8945	AlphaPose	0.5223	0.6402
Pedestrian	GATraj	1.0340	1.8032	OpenPifPaf	0.5187	0.6287
	KeyJoint-STGCN	0.9215	1.6849	–	–	–

The impact of pose trackers is evaluated using PoseFlow, AlphaPose, and OpenPifPaf. As shown in Table 2, OpenPifPaf provides the best trajectory prediction accuracy due to its stable semantic keypoint tracking capability.

3.5 Effect of Input Frames and Empirical Complexity Metrics

The influence of temporal observation length is analyzed by varying the number of past input frames. Table ?? shows that 10 observed frames provide the best trade-off between trajectory prediction accuracy and computational complexity.

Table 3: Ablation study on graph topology.

Graph Topology	JAAD ADE↓	Observation
Single Spatial Graph	0.5912	Baseline
Proposed Multigraph	0.5187	Improved performance

Table 4: Model complexity.

Params (M)	FPS	Input
2.8	21.6	RGB (640 × 480)

The qualitative results demonstrated in Fig 5 that the proposed KeyJoint-STGCN effectively captures articulated motion dynamics and heterogeneous interactions, resulting in improved monocular trajectory prediction performance in complex traffic environments. Table 4 reports the model complexity of the proposed framework, consisting of 2.8M parameters and achieving a real-time inference speed of 21.6 FPS at an input resolution of 640 × 480. An ablation study isolating the graph construction strategy demonstrates that replacing the proposed multigraph with a single spatial graph increases the JAAD Average Displacement Error (ADE) from 0.5187 to 0.5912, highlighting the effectiveness of the multigraph topology in improving trajectory prediction performance. The lightweight monocular design enables real-time trajectory prediction without requiring computationally expensive multi-sensor fusion.

3.6 Discussion Insights

The results on JAAD, nuScenes, and BDD100K show that KeyJoint-STGCN consistently outperforms DESIRE, Semantic-STGCNN, and GATraj in ADE, FDE, and MSE, demonstrating strong cross-dataset general-

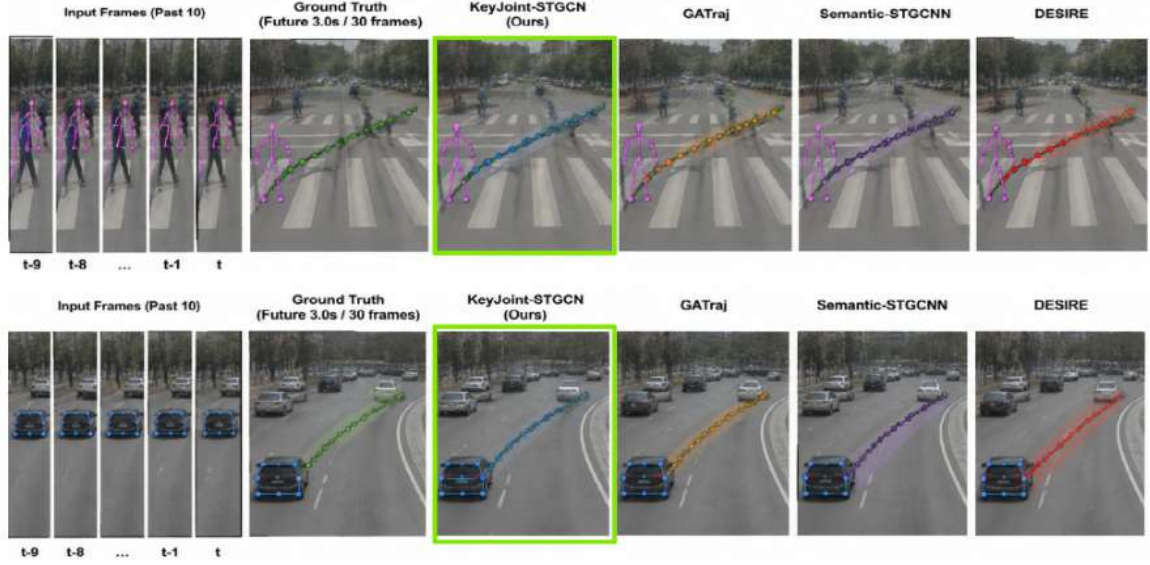


Figure 5: Predicted versus ground-truth (Row1 Fig: Pedestrian using JAAD and Row2 Fig: Vehicle using NuScenes Dataset) trajectory overlays compared to KeyJoint-STGCN marked in green.

ization. This gain is mainly due to keypoint-level motion modeling, which captures fine-grained articulated dynamics beyond bounding box representations.

Pose-aware modeling significantly improves pedestrian prediction by capturing subtle motion cues. Ablation results indicate that 10 input frames offer the best balance between accuracy and efficiency, while OpenPifPaf provides the most stable keypoint tracking, enhancing overall performance.

4 Conclusion and Future Scope

This paper introduced *KeyJoint-STGCN*, a keypoint-based spatio-temporal graph framework for monocular trajectory prediction of pedestrians and vehicles. By leveraging semantic keypoints from OpenPifPaf and a multigraph-STGCN formulation, the method captures fine-grained motion and inter-agent interactions more effectively than object-centric approaches, while operating efficiently on single-camera input. Experiments on JAAD, nuScenes, and BDD100K demonstrate consistent performance improvements and strong generalization across diverse traffic scenarios. However, performance depends on pose estimation quality, which may degrade under occlusion or poor visibility.

Future work will focus on improving robustness via uncertainty-aware or transformer-based pose modeling and incorporating scene context for enhanced prediction in complex environments.

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